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Studierichting: Informatica
Oefeningenreeks 1D nr. 2.4

$$a(z^2 + i) + b(z - 1 - 2i) + c(iz^2 - iz + 3) = (3 + 2i)z - 4 - i$$

$$az^2 + ai + bz - b - 2bi + ciz^2 - ciz + 3c = (3 + 2i)z - 4 - i$$

$$(a + ci)z^2 + (b - ci)z + (ai - b - 2bi + 3c) = (3 + 2i)z - 4 - i$$

$$\begin{cases} a + ci = 0 \Rightarrow a = -ci \\ b - ci = 3 + 2i \Rightarrow b = 3 + 2i + ci \\ ai - b - 2bi + 3c = -4 - i \end{cases}$$

$$(-ci)i - (3 + 2i + ci) - 2(3 + 2i + ci)i + 3c = -4 - i$$

$$-ci^2 - 3 - 2i - ci - 6i - 4i^2 - 2ci^2 + 3c = -4 - i$$

$$c - 3 - 8i - ci + 4 + 2c + 3c = -4 - i$$

$$6c - ci + 1 - 8i = -4 - i$$

$$c(6 - i) = -5 + 7i$$

$$c = \frac{-5 + 7i}{6 - i} = \frac{(-5 + 7i)(6 + i)}{(6 - i)(6 + i)}$$

$$c = \frac{-30 - 5i + 42i + 7i^2}{36 - i^2} = \frac{-37 + 37i}{37} = -1 + i$$

$$a = -(-1 + i)i = i - i^2 = 1 + i$$

$$b = 3 + 2i + (-1 + i)i = 3 + 2i - i + i^2 = 2 + i$$

$$\textbf{Antwoord: } a = 1 + i, b = 2 + i, c = -1 + i$$

References